



Hepatic and Biliary Fascioliasis — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions**1. What is the common name of Fasciola hepatica?**

EASY

- A. Pork tapeworm
- B. Beef tapeworm
- C. Lung fluke
- D. Blood fluke
- E. Sheep liver fluke

2. Where does the adult Fasciola hepatica reside in the definitive host?

EASY

- A. Small intestine lumen
- B. Lungs
- C. Liver and biliary passages
- D. Urinary bladder
- E. Large intestine

3. What is the infective stage of Fasciola hepatica for humans?

EASY

- A. Miracidia
- B. Metacercariae
- C. Cercariae
- D. Sporocyst
- E. Redia

4. Which drug is the treatment of choice for fascioliasis?

EASY

- A. Albendazole
- B. Triclabendazole
- C. Ivermectin
- D. Doxycycline
- E. Metronidazole

5. A patient with fascioliasis has fever, right upper quadrant pain and eosinophilia during the first weeks of infection. Which phase is this?

MEDIUM

- A. Chronic (biliary) phase
- B. Halzoun
- C. Acute (migratory) phase
- D. Convalescent phase
- E. Prepatent latent phase

6. Approximately how long does it take flukes residing in the bile ducts to mature into adult worms?

MEDIUM

- A. 24–48 hours
- B. 1–2 weeks
- C. 1 year
- D. 3–4 months
- E. 5 years

7. Which complication is associated with the chronic biliary phase of fascioliasis?

MEDIUM

- A. Ground itch
- B. Pruritus ani
- C. Hemolytic uremic syndrome
- D. Obstructive jaundice and biliary cirrhosis
- E. Guillain-Barré syndrome

8. Which organism serves as the first intermediate host of *Fasciola hepatica*?

MEDIUM

- A. Aquatic plant
- B. Cow
- C. Pig
- D. Mosquito
- E. Snail

9. A patient develops halzoun after eating raw sheep liver. What is the underlying mechanism?

MEDIUM

- A. Larvae migrate through the lungs
- B. Adult flukes attach to the pharyngeal mucosa, causing edematous congestion
- C. Eggs lodge in the gallbladder
- D. Cercariae penetrate the skin
- E. Metacercariae encyst in muscle

10. Which serologic test becomes positive within about 2 weeks of *Fasciola* infection?

MEDIUM

- A. ELISA
- B. Widal test
- C. VDRL
- D. Weil-Felix test
- E. Rapid plasma reagin

11. Which procedure may be needed to manually extract adult flukes causing biliary obstruction?

MEDIUM

- A. Colonoscopy
- B. Bronchoscopy
- C. ERCP (endoscopic retrograde cholangiopancreatography)
- D. Laparoscopic appendectomy
- E. Upper GI endoscopy alone

12. Which food should be avoided to prevent fascioliasis in endemic areas?

MEDIUM

- A. Undercooked pork
- B. Unpasteurized milk
- C. Raw shellfish
- D. Improperly cooked rice
- E. Raw watercress and other freshwater aquatic plants

13. Approximately how long do adult *Fasciola hepatica* worms live in a human host?

MEDIUM

- A. About 10 years
- B. 2–3 weeks
- C. 6 months
- D. 1 year
- E. 25 years

14. Which measure targets the intermediate host to reduce fascioliasis transmission?

MEDIUM

- A. Vaccination of livestock
- B. Mosquito netting
- C. Chlorination of drinking water only
- D. Deworming of dogs
- E. Snail control

15. A farmer from a sheep-rearing area has tender hepatomegaly and eosinophilia that resolve over several weeks, followed months later by biliary colic and jaundice. Which sequence best explains this course?

HARD

- A. Acute hepatic migration followed by chronic biliary residence and obstruction as the fluke matures
- B. A single acute illness with no further complications
- C. Halzoun followed by spontaneous cure
- D. Chronic phase preceding an acute allergic reaction
- E. Immediate biliary obstruction with no migratory phase

16. Why are eggs typically absent from stool early in fascioliasis despite tissue invasion?

HARD

- A. Eggs are destroyed by gastric acid
- B. Egg-laying only begins once flukes mature in the bile ducts, which takes 3–4 months
- C. The parasite does not produce eggs in humans
- D. Eggs are only found in sputum
- E. Diagnostic PCR is required from day one

17. Both eating raw liver and eating raw watercress can transmit Fasciola-related disease, but via different mechanisms. Which pairing is correct?

HARD

- A. Raw liver → true infection; raw watercress → halzoun
- B. Both cause halzoun only
- C. Both cause true infection only
- D. Raw liver → halzoun (adult worm pharyngeal attachment); raw watercress → true infection (metacercariae ingestion)
- E. Neither route causes human disease

18. Why is aspirated duodenal bile sometimes used instead of stool for diagnosis?

HARD

- A. It may reveal eggs directly from the biliary tract even when stool examination is negative or equivocal
- B. It is the only specimen type accepted for ELISA
- C. Bile aspiration is required before triclabendazole can be prescribed
- D. Stool eggs are never diagnostic for Fasciola
- E. Bile aspiration replaces the need for imaging

19. A patient treated with triclabendazole for fascioliasis later develops biliary colic due to a large dead fluke obstructing the common bile duct. What is the appropriate next step?

HARD

- A. Repeat triclabendazole at a higher dose only
- B. Switch immediately to ELISA-based monitoring only
- C. ERCP for manual extraction of the fluke
- D. Observation without intervention
- E. Start antituberculous therapy

20. A public health team wants to interrupt Fasciola transmission at the point where the parasite multiplies asexually before becoming free-swimming. At which host should control efforts target this stage?

HARD

- A. Aquatic vegetation, where metacercariae encyst
- B. The sheep, where adult worms release eggs
- C. The human, where larvae migrate through the liver
- D. The snail (first intermediate host), where sporocyst, redia and cercaria stages develop
- E. Water sources, where eggs hatch into miracidia

✓ Answer key

Q	Answer	Difficulty	Why
1	E	EASY	<i>F. hepatica</i> primarily infects sheep and cattle, giving it the name sheep liver fluke.
2	C	EASY	The adult fluke lives in the biliary tract after migrating through the liver.
3	B	EASY	Metacercariae encysted on aquatic vegetation are ingested by the definitive host.
4	B	EASY	Oral triclabendazole is specifically named as first-line treatment.
5	C	MEDIUM	This matches the acute phase caused by larval migration through the liver.
6	D	MEDIUM	The chronic phase involves maturation over about 3–4 months in the bile ducts.
7	D	MEDIUM	Chronic biliary residence can cause obstruction, cirrhosis, jaundice, cholelithiasis and anemia.
8	E	MEDIUM	The snail is the first intermediate host; aquatic vegetation is the second.
9	B	MEDIUM	Pharyngeal attachment of adult worms from raw liver causes halzoun.
10	A	MEDIUM	ELISA is the serodiagnostic test noted to become positive within 2 weeks.
11	C	MEDIUM	ERCP allows extraction of flukes causing physical blockage of the biliary tract.
12	E	MEDIUM	Metacercariae encyst on aquatic vegetation such as watercress.
13	A	MEDIUM	Adults live about 5 years in sheep and about 10 years in humans.
14	E	MEDIUM	Controlling snail populations interrupts the life cycle at the first intermediate host.
15	A	HARD	This reflects the natural two-phase course: migratory injury, then biliary maturation and obstruction.
16	B	HARD	Since maturation and oviposition happen in the bile ducts over months, early acute-phase stool exams may be falsely negative.
17	D	HARD	Halzoun follows ingesting adult worms in raw liver; true infection follows ingesting metacercariae on vegetation.
18	A	HARD	Duodenal bile aspiration is listed alongside stool as a method to demonstrate eggs, useful when biliary drainage is sampled directly.
19	C	HARD	Physical blockage of the biliary tract from flukes may require ERCP for manual extraction, per the lecture's management section.
20	D	HARD	Asexual multiplication (sporocyst to redia to cercaria) occurs inside the snail, making snail control a key interruption point.